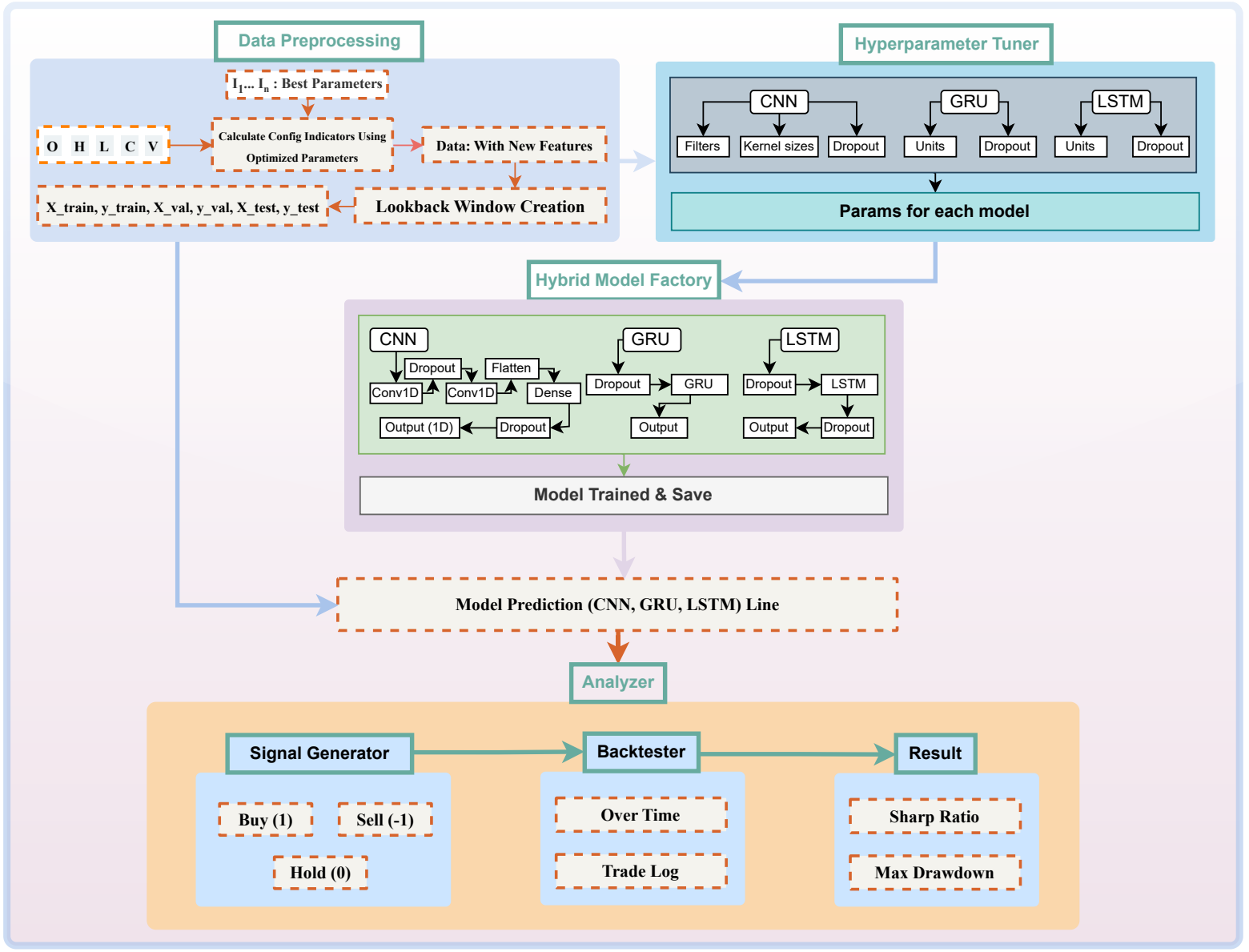

Data Preprocessing
$I_1 ... I_n$: Best Parameters
O H L C V
Calculate Config Indicators Using Optimized Parameters
Data: With New Features
X_train, y_train, X_val, y_val, X_test, y_test
Lookback Window Creation
Hyperparameter Tuner
CNN
Filters
Kernel sizes
Dropout
GRU
Units
Dropout
LSTM
Units
Dropout
Params for each model
Hybrid Model Factory
CNN
Dropout
Flatten
Conv1D
Conv1D
Dense
Output (1D)
Dropout
GRU
Dropout
GRU
Output
LSTM
Dropout
LSTM
Output
Dropout
Model Trained & Save
Model Prediction (CNN, GRU, LSTM) Line
Analyzer
Signal Generator
Buy (1)
Sell (-1)
Hold (0)
Backtester
Over Time
Trade Log
Result
Sharp Ratio
Max Drawdown